CI360 Productivity Intelligence Suite

Complete Technical Architecture & System Documentation

Version: 2.0.0

Environment: Node.js (v18+) · Express.js · MongoDB (Mongoose) · Vite · Vanilla ES Modules

Architecture: Monorepo (Express API Backend + Vite Web Client)

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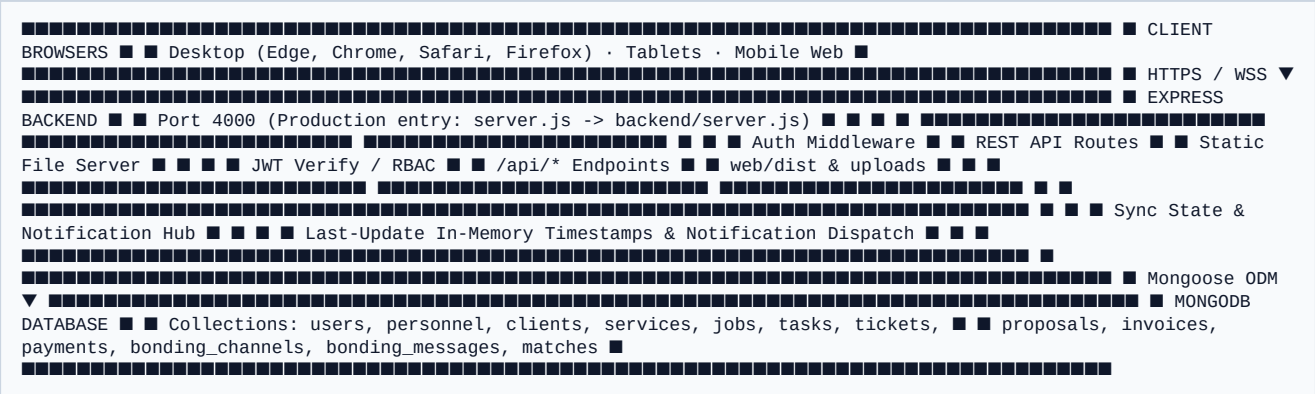
1. System Architecture Overview

CI360 operates as a high-performance, single-instance or distributed Node.js/Express service coupled with a pre-compiled Vite front-end. It provides role-based workspaces for **Super Admins**, **Admins**, **Employees**, and **Clients**.



Directory Structure

```
CIJOB/ci360/ ■■■ package.json # Root scripts (build, dev, start, seed) ■■■ server.js # Production entry point ■■■ docs/ # Technical specs and user manuals ■■■ backend/ # Node.js + Express API Server ■ ■■■ config/db.js # MongoDB connection with Mongoose ■ ■■■ middleware/auth.js # JWT verification & role authorization ■ ■■■ models/ # 21 Mongoose Data Schemas ■ ■■■ routes/ # 18 Modular Express Routers ■ ■■■ services/ # Business logic services ■ ■■■ utils/ # Notification, time & math helpers ■ ■■■ uploads/ # Ticket & invoice attachment storage ■■■ web/ # Modern Frontend (Vite + Vanilla JS + CSS) ■■■ vite.config.js # Build pipeline & development proxy ■■■ index.html # Role router landing page ■■■ login.html # Multi-tenant authentication screen ■■■ admin.html # Superadmin & Admin Workspace ■■■ employee.html # Employee Workspace ■■■ client.html # Client Extranet Portal ■■■ accounts.html # Financial Invoicing Dashboard ■■■ proposals.html # Proposal Studio Workspace ■■■ proposal.html # Client Proposal Review & Signoff ■■■ src/ ■ ■■■ css/ # style.css, tokens.css, supporttickets.css ■ ■■■ js/ # api.js, admin.js, employee.js, planner.js, ■ # supporttickets.js, proposals.js, teambonding.js ■■■ dist/ # Optimized production bundle
```

2. Database Schema & Data Models

The system maintains 21 core collections in MongoDB:

2.1 Core Identity & Operations

- **`User`**: System logins, email, bcrypt password hash, role (**`superadmin`**, **`admin`**, **`employee`**, **`client`**), status flag (**`active`**), personnel reference.
- **`Personnel`**: Staff directory, duties, employment dates, monthly hour capacity (default 160h/month), department assignment.
- **`Client`**: Client brand names, contact details, account managers, retainers, billing profiles, and project links.
- **`Service`**: Agency service offerings (Strategy, CS, Website, Design, Copy, Edit, Shoot, SEO, SMO, QC) with standard rates.
- **`Job`**: Work records tracking title, client ID, service IDs, date, completion status, value, and multi-employee percentage/hour splits.
- **`Roster`**: Staffing matrix mapping personnel to clients and ongoing assignments.
- **`SalaryGrade`** & **`SalaryAssignment`**: Confidential tier grades, pay scales, and personnel compensations (restricted strictly to Super Admin).

2.2 Task, Ticket & Project Workflow

- **`Task`**: Daily Planner tasks with title, priority (**`Low`**, **`Medium`**, **`High`**, **`Urgent`**), status (**`Pending`**, **`In Progress`**, **`Review`**, **`Completed`**), subtask checklist items, and assignees.
- **`SupportTicket`**: Scrum ticket system supporting categories (**`Bug`**, **`Feature`**, **`Task`**, **`Internal`**, **`Infrastructure`**, **`Client Request`**, **`General`**), statuses (**`To Do`**, **`Content`**, **`Creative`**, **`Under Review`**, **`Ready to Leave`**, **`In Progress`**, **`Resolved`**, **`Closed`**), priority levels, rich attachments, reporters, and linked issues (**`relates to`**, **`blocks`**, **`is blocked by`**, **`duplicates`**).
- **`Target`**: Monthly/quarterly revenue and productivity goals assigned to personnel and teams.

2.3 Sales & Billing Subsystem

- **`Proposal`**: Client proposals containing company info, scope items, deliverable milestones, pricing calculations, status (**`draft`**, **`sent`**, **`viewed`**, **`accepted`**, **`declined`**), and e-signature records.
- **`ProposalRateCard`**: Configurable service rate benchmarks.
- **`Invoice`**: Client invoices with invoice numbers, line items, CGST/SGST/IGST tax computations, payment status (**`Draft`**, **`Sent`**, **`Paid`**, **`Overdue`**, **`Cancelled`**), due dates, and PDF exports.
- **`Payment`**: Financial records tracking payment amounts, dates, bank transaction IDs, and methods.
- **`BillingProfile`**: Legal entity details, GSTIN, PAN, bank account coordinates, and payment instructions.

2.4 Team Lounge & Multiplayer Games

- **`BondingChannel`**: Communication rooms with **`type: 'direct'`** (1-on-1 private chat) or **`type: 'channel'`** (group/team lounge), membership arrays, and latest message snapshots.
- **`BondingMessage`**: Individual messages supporting text, emojis, attachments, system alerts, and interactive game challenge cards (**`type: 'challenge'`**).
- **`GameInvitation`**: Match invites specifying **`gameType`** (**`chess`**, **`tictactoe`**, **`ludo`**), sender, recipient, match settings, status (**`pending`**, **`accepted`**, **`declined`**, **`cancelled`**), and target private channel ID.
- **`GameMatch`**: Active match state storing player allocations, turn state, move logs, tournament clocks, and game-specific payloads (Chess FEN / Ludo 15×15 pawn matrices).
- **`BondingScore`**: Employee gamification scores, points awarded for challenges, wins, and daily interactions.
- **`Notification`**: System notifications for tickets, chat mentions, game challenges, and approval alerts.

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3. REST API Endpoints Reference

All API routes are prefixed with **`/api`**. Protected routes require a valid **`Bearer <token>`** HTTP header.

3.1 Authentication (`/api/auth`)

Method	Endpoint	Access	Purpose
`POST`	**`/api/auth/login`**	Public	Validates credentials; returns JWT session token (12h) & user payload
`GET`	**`/api/auth/me`**	Authenticated	Fetches current user profile and role
`POST`	**`/api/auth/change-password`**	Authenticated	Updates current user's password
`POST`	**`/api/auth/reset-password`**	Super Admin	Resets a specific employee's password

3.2 Analytics & Productivity (`/api/dashboard`, `/api/jobs`)

Method	Endpoint	Access	Purpose
`GET`	**`/api/dashboard/admin`**	Admin / Super Admin	Company-wide revenue, utilization, profit margin, capacity warnings
`GET`	**`/api/dashboard/employee`**	Employee	Personal logged hours, targets, utilization percentage
`GET`	**`/api/jobs`**	Authenticated	Lists jobs with period, client, and personnel filters
`POST`	**`/api/jobs`**	Authenticated	Logs a new job with percentage and hour splits
`PUT`	**`/api/jobs/:id`**	Authenticated	Updates existing job
`DELETE`	**`/api/jobs/:id`**	Super Admin / Creator	Deletes a job record

3.3 Daily Tasks & Support Tickets (`/api/tasks`, `/api/tickets`)

Method	Endpoint	Access	Purpose
`GET`	`/api/tasks`	Authenticated	Retrieves daily planner tasks with status filters
`POST`	`/api/tasks`	Authenticated	Creates a new task
`PUT`	`/api/tasks/:id`	Authenticated	Updates task status, checklist, or priority
`GET`	`/api/tickets`	Authenticated	Retrieves support tickets for Scrum board
`POST`	`/api/tickets`	Authenticated	Submits a new support ticket with attachments
`PUT`	`/api/tickets/:id`	Authenticated	Updates ticket status, assignees, priority, comments
`DELETE`	`/api/tickets/:id`	Creator / Superadmin	Strictly deletes a ticket (guarded by creator ID or superadmin role)

3.4 Proposals & Billing (`/api/proposals`, `/api/accounts`)

Method	Endpoint	Access	Purpose
`GET`	`/api/proposals`	Authenticated	Lists client proposals
`POST`	`/api/proposals`	Admin / Super Admin	Creates a proposal draft
`GET`	`/api/proposals/public/:token`	Public	Client access to view and sign proposal
`POST`	`/api/proposals/public/:token/sign`	Public	Records client e-signature and accepts proposal
`GET`	`/api/accounts/invoices`	Admin / Super Admin	Invoicing list with payment tracking
`POST`	`/api/accounts/invoices`	Admin / Super Admin	Generates invoice draft with tax calculations
`POST`	`/api/accounts/payments`	Admin / Super Admin	Records client payment receipts

3.5 Team Bonding & Games (`/api/bonding`)

Method	Endpoint	Access	Purpose
`GET`	`/api/bonding/channels`	Authenticated	Lists available 1-on-1 direct chats and group channels
`POST`	`/api/bonding/channels`	Authenticated	Creates group channel or direct message thread
`GET`	`/api/bonding/channels/:id/messages`	Authenticated	Loads message stream with pagination
`POST`	`/api/bonding/channels/:id/messages`	Authenticated	Posts message or media attachment
`POST`	`/api/bonding/invitations`	Authenticated	**Creates a game challenge delivered strictly to personal direct chat**
`GET`	`/api/bonding/invitations`	Authenticated	Lists pending incoming and sent challenges
`PUT`	`/api/bonding/invitations/:id/respond`	Authenticated	Accepts or declines challenge
`POST`	`/api/bonding/matches/instant-ludo`	Authenticated	Launches 2P, 3P, or 4P Ludo match
`GET`	`/api/bonding/matches/:id`	Authenticated	Fetches live game board state
`POST`	`/api/bonding/matches/:id/move`	Authenticated	Submits move or rolls 3D dice (verified turn)
`DELETE`	`/api/bonding/matches/:id`	Players / Admin	Cleans up and archives game match record

4. Authentication & Authorization Framework

4.1 JWT Token Handling

Authentication is implemented via JSON Web Tokens (JWT). Upon successful credentials validation:

1. The server signs a payload containing `{ id, role, email, name }` using `JWT_SECRET` with a 12-hour expiration.
2. The client stores this token in `localStorage.getItem('ci360_token')`.
3. Every API call attaches the header: `Authorization: Bearer <token>`.
4. Token expiration or `401 Unauthorized` responses automatically trigger redirection to `/login.html`.

4.2 Role-Based Access Control (RBAC) Matrix

Feature / Module	Super Admin	Admin	Employee	Client
Company Financial Dashboard	■ Full	■ Full	■ Blocked	■ Blocked
Personal Utilization & Jobs	■ Full	■ Full	■ Self Only	■ Blocked
Log & Edit Jobs	■ All	■ All	■ Self Only	■ Blocked
Client & Service Directory	■ Manage	■ Manage	■■ Read	■ Blocked
Salaries & Compensation	■ Full	■ Blocked	■ Blocked	■ Blocked
User Accounts & Access	■ Manage	■ Blocked	■ Blocked	■ Blocked
Support Tickets (Create/Edit)	■ Full	■ Full	■ Full	■ Blocked
Support Tickets (Delete)	■ Full	■ Blocked	■ Creator Only	■ Blocked
Proposal Studio & Signoffs	■ Full	■ Full	■■ Read	■ Client Review
Invoicing & GST Accounts	■ Full	■ Full	■ Blocked	■ Blocked
Team Chat & Games Arena	■ Full	■ Full	■ Full	■ Blocked

5. Real-time Polling & Synchronization Engine

To eliminate server overhead and maintain resilient connections across firewalls, CI360 uses an **Event-Driven Synchronization Hub**:

1. **In-Memory Timestamp Counters**:

The backend tracks global modification timestamps:

- `__ci360LastGlobalUpdate`
- `__ci360LastTicketUpdate`
- `__ci360LastTaskUpdate`
- `__ci360LastInvoiceUpdate`
- `__ci360LastChatUpdate`
- `__ci360LastBondingUpdate`

2. **Lightweight Heartbeat Polling**:

Clients ping `/api/notifications/sync-status` every 3 seconds:

- If timestamps match the client's cached state, the server returns `{ modified: false }` with zero database overhead.
- If an entity was updated (e.g., ticket moved, dice rolled, chat sent), the client re-fetches only the affected view without re-rendering the whole page.

6. Design System, Theming & Layout Spacing Standard

6.1 Theme Tokens & Dark Mode

CI360 supports **Light Theme** (default) and **Dark Theme** via CSS custom variables (`[data-theme="dark"]`). Tokens adapt typography, borders, surfaces, and shadows seamlessly across all screens.

6.2 Global Content Spacing Standard

All primary views inside the application shell are governed by an explicit CSS container standard in `style.css`:

```
.app-content { max-width: none !important; width: auto !important; box-sizing: border-box !important; flex: 1 1 0%;
padding: 20px !important; margin: 0px 71px !important; } @media (max-width: 992px) { .app-content { margin: 0 32px
!important; padding: 20px !important; } } @media (max-width: 768px) { .app-content { margin: 0 16px !important;
padding: 16px !important; } }
```

- **Uniform 20px Padding**: Applied inside the main viewport on top, right, bottom, and left.
- **71px Horizontal Desktop Margin**: Establishes a comfortable reading margin on widescreen desktop displays.
- **Responsive Fallback**: Gracefully scales to 32px on tablets and 16px on mobile viewports.

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7. Subsystem Architecture

7.1 Executive Dashboard & Capacity Intelligence

Aggregates billable hours against employee capacity (default 160 hrs/mo) and calculates workload utilization percentages:

$$\text{Utilization \%} = \left(\frac{\text{Logged Billable Hours}}{\text{Monthly Target Hours}}\right) \times 100$$

- Highlights over-utilized (>100%) and under-utilized (<70%) team members with proactive capacity redistribution advice.

7.2 Job Logging & Billable Split Engine

Supports multi-person work distribution on a single job:

- Multiple team members can be assigned percentage splits $\sum \% = 100\%$ or individual logged hours.
- Revenue recognition automatically divides client billings across participating personnel.

7.3 Daily Planner & Task Kanban

Features a priority matrix (Urgent, High, Medium, Low), inline task creation, subtask checklist completion percentages, and drag-and-drop or click-to-move status updates.

7.4 Support Tickets & Jira Scrum System

- 5-Column to 8-Column Scrum board with drag/select progression.
- Fast streaming multi-file upload system and external reference URL linking.
- **Strict Security Guard**: Only the original ticket reporter or a Superadmin may delete a ticket.

7.5 Proposal Studio & Rate Card Engine

Interactive proposal generator:

- Scopes deliverables, milestones, payment schedules, and terms.
- Generates secure public client links for digital signature and acceptance.

7.6 Accounts, Invoicing & GST Financials

- Automatically formats invoices compliant with Indian GST (CGST 9% + SGST 9% or IGST 18%).

- Tracks accounts receivable, aging buckets (0-30 days, 31-60 days, 60+ days), and payment logs.

7.7 Team Chat, Lounge & Multiplayer Games Arena

- **Personal Chat Challenge Delivery**: Every challenge generates an invitation strictly into the 1-on-1 direct conversation between the two colleagues.
- **Ludo Royale**: 15×15 grid cross track, 4 pawns per color, auto-advance for single valid moves, safe squares, and a 3D tactile dice with realistic animations and isolated styling.
- **Chess Duel**: Standard FEN validation via `chess.js`, legal move dot indicators, live clocks, and notation.
- **Tic-Tac-Toe**: Turn-based fast match.

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8. Production Build, Deployment & DevOps

8.1 Build Pipeline

The web client is compiled using Vite:

```
cd web npm run build
```

This builds production bundles into `web/dist/`, minifying JavaScript, optimizing stylesheets, and tree-shaking unused libraries.

8.2 Production Server Execution

The Express application serves the API and the compiled frontend simultaneously:

```
# In project root npm run start
```

Runs `node server.js` which loads environment variables and serves `web/dist` on port 4000.

CI360 Productivity Intelligence Suite

Complete End-User & Administrator Operating Manual

Welcome to CI360

CI360 is an enterprise productivity tracker, work logging system, Jira-style support ticket desk, financial invoicing hub, and team bonding lounge designed for digital agencies, consultants, and project teams.

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13. **Troubleshooting & Frequently Asked Questions (FAQ)**

1. Getting Started & Logging In

1.1 Accessing the Platform

1. Open your web browser (Google Chrome, Microsoft Edge, Safari, or Firefox).
2. Navigate to your agency's CI360 URL (e.g. `http://localhost:4000` or your production domain).
3. If not already signed in, you will be directed to the login portal.

1.2 Signing In

1. Enter your registered work **Email** and **Password**.
2. Click **Sign In to Workspace**.
3. You will be automatically routed to your assigned workspace based on your role:

- **Super Admin & Admin** ➔ Redirected to `admin.html`
- **Employee** ➔ Redirected to `employee.html`
- **Client** ➔ Redirected to `client.html`

1.3 Theme Toggle & Password Management

- **Switch Themes**: Click the ■■/■ icon in the top-right header to toggle between **Light Theme** and **Dark Theme** instantly.
- **Change Password**: Click your user profile avatar in the upper right, select **Settings / Change Password**, enter your current password, your new password, and click **Update Password**.

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2. Interface Overview & Global Controls

■■■

3.3 Capacity & Burnout Warnings

The dashboard monitors employee capacity (160 hours/month default):

- ■ **Over-Capacity (>100%)**: Flagged in red with actionable guidance to reassign workloads or bring in freelance support.
- ■ **Optimal (80% - 100%)**: Peak team efficiency.
- ■ **Available Bandwidth (<75%)**: Ready for new job assignments.

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4. Module 2: Employee Workspace & Workload Tracking

Employees see a dedicated, focused workspace highlighting their daily responsibilities, hours, and targets.

4.1 Utilization Gauge

Displays your current month's logged hours against your target capacity (e.g. `124h / 160h logged (78%)`).

4.2 Lead Management (Mansi & Urna Lead Workspaces)

Special lead personnel (e.g. Mansi & Urna) have team oversight views enabling them to review duties, reassign subtasks, and track department output.

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5. Module 3: Daily Planner & Task Management

The Daily Planner gives team members a clear personal checklist and team Kanban board.

5.1 Creating a Daily Task

1. Navigate to **Daily Planner** from the sidebar.
2. In the **New Task** input box, enter your task title.
3. Select **Priority** (`Low`, `Medium`, `High`, `Urgent`).
4. Set an optional **Due Date** and **Assignee**.
5. Click **Add Task**.

5.2 Task Progress & Checklists

- **Checklist Subtasks**: Click on any task to add itemized checklists. As items are checked off, a progress bar automatically calculates the completion percentage.
- **Status Updates**: Click or drag tasks between **Pending**, **In Progress**, **Under Review**, and **Completed**.

---

6. Module 4: Job Logging & Time Tracking

Job logging records client work and divides credit accurately among participating employees.

6.1 How to Log a Job

1. Click **+ Log Job** in the topbar or navigate to **Log a Job** in the sidebar.
2. Enter the **Job Title / Deliverable Description**.
3. Select the **Client** from the dropdown menu.
4. Select one or more **Services** applied (e.g. `Design`, `Copy`, `SEO`, `Edit`).

5. Choose the **Job Date**.
6. Enter the **Total Job Value (■)** or hourly billing rate.
7. **Allocate Multi-Person Work Split**:
- Assign the primary employee (e.g. `Person 1: 70%`).
- Assign the supporting colleague (e.g. `Person 2: 30%`).
- The system automatically validates that total splits equal 100% and calculates logged hours accordingly.
8. Click **Save Job Record**.

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7. Module 5: Support Tickets (Jira Scrum Board)

A complete Jira-style issue tracking and Scrum management desk.

7.1 The Scrum Columns

Tickets flow across standard lifecycle stages:

1. **To Do**: Backlog tickets awaiting initiation.
2. **Content**: Copywriting and briefing stage.
3. **Creative**: Design, video editing, and visual asset preparation.
4. **Under Review**: Internal quality check and client feedback.
5. **Ready to Leave**: Approved and queued for publication or deployment.
6. **Resolved / Closed**: Completed tickets.

7.2 Submitting a Support Ticket

1. Click **Create Ticket** on the Scrum toolbar.
2. Enter a descriptive **Title** (e.g. `SUP-104: Fix checkout currency rounding`).
3. Select **Category** (`Bug`, `Feature`, `Task`, `Internal`, `Infrastructure`, `Client Request`).
4. Set **Priority** (`Low`, `Medium`, `High`, `Critical`).
5. Assign one or more **Assignees**.
6. Attach screenshots or files using the fast parallel streaming file uploader or paste external links (Figma, Google Drive, GitHub PRs).
7. Click **Create Ticket**.

7.3 Ticket Deletion Security Policy

> ■ ***Security Notice**: To prevent accidental data loss and maintain accountability, **only the user who originally created the ticket OR a Superadmin can delete it**. Other users will see deletion options restricted with a clear permission notice.*

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8. Module 6: Proposal Studio & Client Signoffs

The Proposal Studio enables rapid generation of commercial proposals and contract agreements.

8.1 Creating a Client Proposal

1. Open **Proposal Studio** from the sidebar.
2. Click **Create Proposal**.
3. Choose the client, project title, and validity period.
4. Add **Deliverables & Milestones**:

- Item title, description, estimated timeline, and commercial fee.

5. Review total project valuation and payment schedule terms (e.g. `50% Advance, 50% on Delivery`).
6. Click **Generate Proposal**.

8.2 Client Review & E-Signature Portal

1. Click **Share Link** to copy the secure, tokenized client URL (`/proposal.html?token=...`).
2. Send the link to the client.
3. When the client opens the link:

- They review the detailed scope of work.
- They enter their full legal name and type/draw their signature.
- Upon clicking **Accept & Sign Proposal**, the proposal status flips to **Accepted**, triggering an instant notification to the admin team.

---

9. Module 7: Accounts, Invoicing & GST Billing

Manage client invoices, accounts receivable, and Indian GST tax calculations.

9.1 Generating an Invoice

1. Open **Accounts & Billing** from the sidebar (or navigate to `/accounts`).
2. Click **New Invoice**.
3. Select the **Client**. The system automatically pulls their GSTIN, address, and legal billing profile.
4. Add line items, quantities, and unit rates.
5. Select tax configuration:

- **Intra-State**: CGST (9%) + SGST (9%)
- **Inter-State**: IGST (18%)

6. The subtotal, tax amounts, and grand total are calculated automatically.
7. Click **Save as Draft** or **Issue & Send**.

9.2 Recording Payments

1. Locate the invoice under **Invoices Roster**.
2. Click **Record Payment**.
3. Enter the amount received, payment date, payment mode (`NEFT`, `RTGS`, `UPI`, `Cheque`), and transaction reference ID.
4. The invoice status updates to **Paid** or **Partially Paid**.

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10. Module 8: Team Chat, Lounge & Multiplayer Games Arena

A workplace hub for team communication and friendly competition.

10.1 Direct 1-on-1 Chats vs. Group Lounges

- **1-on-1 Direct Chats**: Private end-to-end messaging between any two colleagues.
- **Group Lounges**: Topic channels (e.g. `#general`, `#design-team`, `#coffee-break`) for group discussion.

10.2 Game Challenge Personal Chat Guarantee

> ■■ ***Privacy Rule**: All game challenges are **strictly delivered to the opponent's personal 1-on-1 direct chat**, never into public channels or group rooms.*

Sending a Challenge:

1. Click ■■ `Challenge` in the top action toolbar, or click `Play` in the Games Lounge.
2. Select the game: **Ludo Royale**, **Chess Duel**, or **Tic-Tac-Toe**.
3. For Ludo: Choose **2 Players**, **3 Players**, or **4 Players**.
4. Select your opponent(s).
5. Click **Send Challenge** ■.
6. The challenge card is delivered directly into your opponent's personal chat thread.

Accepting a Challenge:

1. The opponent receives a notification and sees a ■■ `Game Challenge` tag on their direct chat list.
2. They open the private chat and click ■■ `Accept Challenge`.
3. The live match arena modal opens automatically on both screens.

10.3 Ludo Royale: Rules & 3D Tactile Dice

- **15×15 Track**: 4 pawns per player. Pawns release from Base Yard (`step: -1`) to the track on rolling a **6**.
- **Safe Squares (■)**: Squares marked with stars protect your pawns from being captured.
- **Home Triumph (■)**: Pawns advance through their colored home stretch into the center crown square `56`.
- **3D Tactile Dice**: Click either the 3D ivory die or the `[■ TAP TO ROLL]` button below it.
- **Auto-Advance**: When only one pawn has a legal move following your roll, the system automatically advances it to keep the game fast and fluid.

10.4 Chess Duel

- Full chess rules powered by `chess.js`.
- Click any piece to view legal move dots (■); tap the destination square to make your move.
- Choose from Untimed or timed Blitz/Rapid clocks.
- Sidebar moves notation records every turn.

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11. Module 9: Settings, Roster, Targets & Salaries (Super Admin)

Super Admins have exclusive administrative tools:

- **Personnel Roster Management**: Add new staff, set monthly hours capacity, and assign departments.
- **Monthly Targets**: Configure revenue targets for individuals and teams.
- **Salary Grades**: Confidential salary bands and compensation management.
- **User Accounts**: Create logins, assign roles, toggle account active states, and perform password resets.

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12. Module 10: Client Extranet Portal

Clients who sign in with client credentials receive a simplified, branded extranet view:

- Review in-progress projects and milestones.
- View deliverable attachments and approved assets.

- Download issued invoices and payment receipts.
- Cannot access internal team chats, salary data, or employee logs.

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13. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: The dice in Ludo was showing a blank square. Is this resolved?

A: Yes. The blank square was caused by a CSS name conflict with a global `.empty` class that had 52px padding. The pip styles have been isolated with `.ludo-pip-filled` and `.ludo-pip-empty`, reset protection, and enhanced 3D tactile cubic styling.

Q2: Why did my game challenge appear in a group room?

A: In earlier versions, sending a challenge while inside a group chat could post the invitation to that channel. The system now strictly intercepts all game challenges and routes them **exclusively into the 1-on-1 personal direct chat** between you and your colleague.

Q3: Why can't I delete a support ticket?

A: To ensure data integrity, support tickets can only be deleted by the person who originally reported/created the ticket, or by a Superadmin.

Q4: How do I change margins or padding across the app?

A: The entire software conforms to the `.app-content` standard: `padding: 20px !important; margin: 0px 71px !important;` on desktop displays, scaling down automatically for mobile and tablet devices.

CI360 Team Lounge & Multiplayer Games Arena

Comprehensive Technical Documentation & User Manual

Executive Overview

The **Team Chat & Lounge (Team Bonding)** module in CI360 combines real-time workplace collaboration with interactive multiplayer gaming. Designed to foster teamwork, healthy workplace competition, and rapid communication, the system integrates 1-on-1 direct messaging, group team lounges, and an interactive games arena featuring **Ludo Royale (2P, 3P, 4P)**, **Chess Duel**, and **Tic-Tac-Toe**.

1. System Architecture & Routing Policy

1.1 Personal Chat Routing Guarantee

A core architectural requirement of CI360 is that **all game challenges are strictly private between the participating colleagues**:

- **Zero Group Contamination**: When a challenge is issued (even from within a group channel or the main Lounge), it is **never** broadcast to public or group rooms.
- **Automatic 1-on-1 Provisioning**: The backend automatically looks up the direct 1-on-1 channel (`type: 'direct'`) between the challenger and the recipient. If one does not already exist, it is created instantly.
- **Challenge Lifecycle in Private Chat**:

1. The interactive challenge card appears solely inside the recipient's personal chat stream.
2. The personal chat conversation is updated with a `challenge` badge and bumped to the top of both users' recent chat lists.
3. Push/in-app notifications direct the recipient directly into the private 1-on-1 conversation.
4. Once accepted, match start notices and game arena launchers are contained strictly within the personal thread.

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[Challenger Issues Invite] ■ ▼ [Backend Intercepts] ■■■ Inspects target colleague ■ ■■■■■ Locates or auto-creates 1-on-1 Direct Chat (`type: 'direct'`) ■■■■■ Inserts interactive challenge card into 1-on-1 chat stream ■■■■■ Dispatches high-priority notification to recipient ■■■■■ [Group Channels & Team Lounges remain completely clean]
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2. Multiplayer Games Arena

2.1 Ludo Royale (2-Player, 3-Player, 4-Player)

An authentic, rule-compliant implementation of classic Ludo designed for cross-office multiplayer sessions.

A. Board Layout & Dimensions

- **15×15 Cross Track**:
- **4 Corner Yards (6×6 each)**: Red (Top-Left), Green (Top-Right), Blue (Bottom-Left), Yellow (Bottom-Right).
- **Safe Squares (■)**: Squares `0, 8, 13, 21, 26, 34, 39, 47` are designated safe zones where pawns cannot be captured.
- **Start Squares (➔)**: Entry squares for released pawns.
- **Home Stretches (5 squares per arm)**: Colored pathways leading directly to the center triangle.

- **Home Triumph Core (■)**: Final square `56`. When all 4 pawns reach this center core, the player wins.

B. 3D Tactile Interactive Dice

The dice UI has been crafted to simulate a real, physical gaming die:

- **Ivory Acrylic Casing**: 64×64px rounded cube (`border-radius: 14px`) with soft top-down lighting and a 6px physical bottom shelf shadow (`box-shadow: 0 6px 0 #cbd5e1`).
- **Dynamic Turn Colors**: The border and ambient glow dynamically change according to whose turn it is:
- ■ **Red**: `#ef4444`
- ■ **Green**: `#10b981`
- ■ **Yellow**: `#f59e0b`
- ■ **Blue**: `#3b82f6`
- **Concave 3D Pips**:
- Deep obsidian pips with radial shading and light reflection.
- **Signature Ace Pip**: Face 1 features a classic enlarged crimson-red center pip.
- **Reset Isolation**: Pips are explicitly isolated (`ludo-pip-filled` / `ludo-pip-empty`) so global page styles cannot distort the 3×3 grid.
- **Physical Feedback & Animations**:
- **Idle / Your Turn**: Energetic `pulse-roll` 3D wobble encouraging the player to roll.
- **Press Feel**: Compresses 3px downward on mouse down or tap, mimicking a tactile mechanical button.
- **Rolling Tumbler**: Smooth 360° rotational tumble animation while the backend evaluates the outcome.
- **Dual Click Action**: Both the 3D dice and the `[■ TAP TO ROLL]` badge underneath are clickable.

C. Gameplay Rules & Logic

1. **Pawn Release**: A roll of **6** is required to move a pawn from its Base Yard (`step: -1`) to the track (`step: 0`).
2. **Track Progression**: Pawns navigate clockwise around the 52 perimeter squares.
3. **Captures**: Landing on an opponent's pawn on a non-safe square captures it, sending it back to their Base Yard (`step: -1`).
4. **Auto-Advance**: If only a single pawn has a valid legal move following a roll, the game automatically advances it without requiring an extra tap, speeding up game pace.

2.2 Chess Duel

- **Legal Validation**: Full chess engine powered by `chess.js`, enforcing legal moves, castling, en passant, pawn promotion, check, checkmate, and stalemate.
- **Interactive Board**: Click piece to reveal green highlighted legal move dots (`.legal-dot`); click destination square to execute move.
- **Time Controls**: Supports Untimed, 1 min (Bullet), 3 min (Blitz), 5 min (Rapid), and 10 min matches.
- **Notation Feed**: Sidebar live notation recording every move in algebraic standard format.

2.3 Tic-Tac-Toe Showdown

- Lightweight 3×3 quick-play game.
- Turn-based alternating moves (`X` vs `O`).
- Instant line-check win detection and tie recognition.

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3. User Manual & Step-by-Step Operator Guide

3.1 How to Challenge a Colleague

1. Navigate to **Team Chat & Lounge** from the left navigation sidebar.
2. You can initiate a challenge in three ways:

- Click the ■■ `Challenge` button in the top action toolbar.
- Click the ■ `Challenge Colleague` button inside any colleague's 1-on-1 chat header.
- Switch to the **Lounge & Games Arena** view and click `Play` on your desired game.

3. In the Challenge Modal:

- **Step 1**: Select the game type (**Ludo Royale**, **Chess Duel**, or **Tic-Tac-Toe**).
- **Step 2 (Ludo Only)**: Choose the player count (**2 Players**, **3 Players**, or **4 Players**).
- **Step 3**: Search or select your opponent(s) from the colleague list.
- **Step 4 (Chess Only)**: Choose time controls (e.g. Untimed or 5 Minutes).

4. Click `Send Challenge ■`.
5. The challenge card is delivered directly into your colleague's personal 1-on-1 chat.

3.2 How to Accept and Join a Match

1. When a colleague challenges you:

- An in-app badge notification appears on your bell icon.
- The colleague's direct conversation will display a ■■ `Game Challenge` tag.

2. Open their personal direct chat.
3. On the interactive Game Challenge Card, click ■■ `Accept Challenge`.
4. The live match arena modal will open immediately on both screens.

3.3 Playing a Match

1. **Rolling the Dice (Ludo)**:

- When the banner shows ■ `It is your turn!`, click either the 3D dice or the ■ `TAP TO ROLL` badge.
- If you roll a 6, tap a pawn in your yard to release it.
- If multiple pawns can move, movable pawns will pulse with a blue glow. Click the desired pawn to advance it.
- If only one pawn can move, it will auto-advance automatically.

2. **Moving Pieces (Chess)**:

- Click on your piece to see highlighted dots on valid destination squares.
- Tap the desired green dot to complete your turn.

3. **Resigning or Concluding**:

- At any time during an active game, click ■■ `Resign` to concede the match gracefully.
- To leave or review the board, click the ✕ close button in the upper-right corner. Matches stay live and can be resumed at any time from the Lounge or personal chat.

4. UI Layout & Global Spacing Standards

The application layout conforms to a strict, standardized spacing hierarchy:

Element	Property	Value	Purpose
Main Content Container	`padding`	`20px !important`	Clean uniform buffer on all sides
Main Content Container	`margin`	`0px 71px !important`	Precise horizontal alignment on desktop
Main Content Container	`flex`	`1 1 0%`	Fluid vertical and horizontal filling
Main Content Container	`max-width`	`none !important`	Prevents cramped container cutoffs
Dashboard & Metric Cards	`padding`	`20px`	Consistent interior padding
Mobile Breakpoint (≤768px)	`margin`	`0px 16px !important`	Responsive buffer for handheld screens
Mobile Breakpoint (≤768px)	`padding`	`16px !important`	Preserves maximum screen real estate

5. Security & Permission Reference

1. **Role Access**:

- **Super Admin & Admin**: Can participate in challenges, create direct and group channels, manage member rosters, and delete concluded matches.
- **Employee**: Can challenge colleagues, participate in all games, manage personal chats, and view leaderboard scores.
- **Client**: Restricted to client dashboard; cannot access the internal employee lounge or games arena.

2. **Turn Authentication**:

- Every move request to `/api/bonding/matches/:id/move` verifies the user's JWT token. Players cannot move opponent pieces, roll out of turn, or submit unauthorized moves.

3. **Data Sanitization**:

- All messages, player tags, and game moves are HTML-escaped to prevent cross-site scripting (XSS).